

Postgres monitoring review checklist

This document describes what every good Postgres monitoring must or should have. The document contains two sections: first, we describe the actual “monitoring” part, and then overview the list of alerts that are worth having.

👉 Follow-up doc: [PostgreSQL Troubleshooting & Monitoring – Checklist / Runbook](#)

Time-series graphs and tables

Color highlighting meanings:

- most important,
 - very good to have,
 - unclassified (yet),
 - new, unclassified
- without highlighting: good to have, may be helpful for troubleshooting, but we can survive without it.

System

- Basic
 - CPU
 - LA
 - Memory usage
 - Memory errors
 - Swap usage
 - Swap I/O
- Disk
 - Read and write latency
 - IOPS (read and write)
 - Throughput (read and write)
 - % util / io usage
 - Disk space
 - Page cache hit/miss
- Network
 - Bandwidth

- Errors, dropped packets
 - Packets/second
 - Connections (in / out; per status / per IP)
- Process analysis
 - Top-N by CPU load
 - Top-N by RAM load
 - Top-N by disk reads (IOPS and throughput)
 - Top-N by disk writes (IOPS and throughput)
 - Top-N by swap (amount + rates)
 - Top-N by process count and by threads count
 - Top-N by open files

Postgres

- Main
 - Average query time
 - Connections (by state in `pg_stat_activity`)
 - Top-N transactions by age
 - Total query time (by query group from `pg_stat_statements`)
 - Buffers hits and reads (by query group from `pg_stat_statements`)
 - Buffers written, dirtied (by query group from `pg_stat_statements`)
 - Transactions (committed vs rolled back)
 - Replication lags
- Autovacuum
 - Workers (by mode: main vs tx ID wraparound)
 - Top-N tables (and DBs) by autovacuum pending work
 - TXs left (tx ID wraparound), database transaction age
 - Autovacuum queue and progress <https://gitlab.com/snippets/1889668>
- Bloat (estimated)
 - Table bloat (estimated)
 - Index bloat (estimated)
- Checkpoints, bgwriter
 - Disk writes (by processes)
 - Pages dirtied by queries
 - Checkpoints issued
- Connections
 - # of connections by state + max_connections
 - Connections by client address (active + all)
 - Connections by app (active + all)
 - Connections by DB user
 - Connections by database name
 - Idle in transaction connections

- Locks, waits
 - Total number of locks acquired
 - Locks by time
 - Queries blocked longer than X seconds
 - Deadlocks
 - Types of waits
- Replication
 - Destination (follower)
 - Replication lag in bytes (for followers of the current node)
 - Replication lag in bytes (for the current node compared to the leader)
 - Replication lag in seconds (for the current node compared to the leader)
 - Origin (primary or replica with cascaded replication)
 - Unused replication slots / replication statuses
 - Amount of bytes for each replication slot
 - Replication lag phases (1 graph for each follower)
- Tables
 - Table sizes (total, heap, TOAST, indexes)
 - Estimated rows
 - Top-N by seqscan
 - Top-N by blocks read
 - Top-N by INSERT
 - Top-N by UPDATE
 - Top-N by DELETE
 - Top-N by size (tuples, bytes)
 - Top-N by bloat (estimated!)
 - Top-N by n_dead_tup
- Indexes
 - Index sizes
 - Index usage
 - Not valid indexes
 - Unused indexes
 - Redundant indexes
- Functions
 - Function usage (calls per second)
 - Average time of execution (total, self)
- WAL
 - pg_xlog/pg_wal size
 - Archiver statuses (fail/success)
 - WAL write rates, B/s
 - WAL files count (total, unarchived)
 - WAL directory size

- WAL files which ready to be archived (count of files in pg_xlog/pg_wal/archive_status which end in ".ready")
- Transactions
 - Transactions per second (TPS)
 - Long-running transactions / max transaction age
- Query macro-analysis based on pg_stat_statements
 - Top-N by total_time
 - Top-N by mean_time
 - Top-N by calls
 - Top-N by CPU usage
 - Top-N by I/O timing
 - Top-N by I/O timing - writes
 - Top-N by block reads (page cache->buffer pool)
 - Top-N by blocks dirtied
 - Top-N by rows
 - Top-N by block hits (buffer pool)
 - Top-N by temporary files generated (bytes; blocks)
 - Top-N by block reads from disk
 - Top-N by block writes
 - --- ability to filter and/or aggregate by dbid ("no filter" also wanted)
 - --- ability to filter and/or aggregate by userid ("no filter" also wanted)
 - --- ability to filter and/or aggregate by "the first word" (SELECT/INSERT/...) ("no filter" also wanted)
 - --- ability to filter and/or aggregate by relations mentioned in query text ("no filter" also wanted)
- For each query group from the top-N list -- personal graphs showing:
 - mean_time
 - total_time (?)
 - calls
 - block operations
 - rows
 - query stages: CPU, I/O read, I/O write
 - more info: pg_stat_kcache, pg_qualstats, pg_sortstats
- Macro-analysis based on wait events
 - Query groups by wait event types
 - Query groups by wait events
 - Top-N by time spent in wait event (agg. by type)
 - Top-N by time spent in wait event
 - For each query
 - History of the query group: wait event types
 - History of the query group: wait events
 - For each wait event type:

- History of query groups
 - For each wait event:
 - History of query groups
 - Time spent in each event withing this type
- Log analysis
 - Critical events: restarts, crashes
 - Autovacuum activity
 - Checkpointer activity
 - Locks (>deadlock_timeout)
 - Deadlocks
 - Query examples
 - Plan examples
 - Connections, disconnections
- pgBouncer monitoring
 - From pgbouncer log:
 - Average query time
 - Average transaction time
 - Queries per second (QPS)
 - Transactions per second (TPS)
 - Traffic in and out, B/s
 - Number of connections by client addr
 - Connections between pgBouncer and Postgres, by state
 - Utilization for each pool
 - Waiting clients and waiting time
- Backups
 - ?
 - Time from last successful time (graph)
 - Last backup size (graph)

Alerts

WIP

Critical

- *Disk space (% or in GiB)*
- *Number of connections is close to max_connections (%)*
- *Number of idle-in-transaction connections > N*
- *Inactive replication slots*
- *Replication slot size is > X*

- *Autovacuum workers = autovacuum_max_workers*
- *Transaction ID wraparound risk*
- *Archives failed X times in Y seconds*
- *Long blocking session (e.g. >1min)*
- *Time from last successful backup (e.g. >30hours)*